Lab 1: Packet analysis at application layer using Wireshark

SCSR1213 Network Communications

Universiti Teknologi Malaysia

Objective:

1. Understanding of network protocols by observing the sequence of messages exchanged between two protocol entities, delving down into the details of protocol operation, and causing protocols to perform certain actions and then observing these actions and their consequences.

2. To introduce student with Wireshark software tool for packet analyzer.

3. To analyze protocol used in application layer such as http and dns.

Reference material: Computer Networking: A Top-Down Approach, 7th ed., J.F. Kurose and K.W. Ross.

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PART A: Wireshark Getting Started

1.0 Introduction

The basic tool for observing the messages exchanged between executing protocol entities is called a **packet sniffer**. As the name suggests, a packet sniffer captures ("sniffs") messages being sent/received from/by your computer; it will also typically store and/or display the contents of the various protocol fields in these captured messages. A packet sniffer itself is passive. It observes messages being sent and received by applications and protocols running on your computer, but never sends packets itself. Similarly, received packets are never explicitly addressed to the packet sniffer. Instead, a packet sniffer receives a *copy* of packets that are sent/received from/by application and protocols executing on your machine.

Figure A.1 shows the structure of a packet sniffer. At the right of Figure 1 are the protocols (in this case, Internet protocols) and applications (such as a web browser or ftp client) that normally run on your computer. The packet sniffer, shown within the dashed rectangle in Figure A.1 is an addition to the usual software in your computer, and consists of two parts. The **packet capture library** receives a copy of every link-layer frame that is sent from or received by your computer. In Figure A.1, the assumed physical media is an Ethernet, and so all upper-layer protocols are eventually encapsulated within an Ethernet frame. Capturing all link-layer frames thus gives you all messages sent/received from/by all protocols and applications executing in your computer.

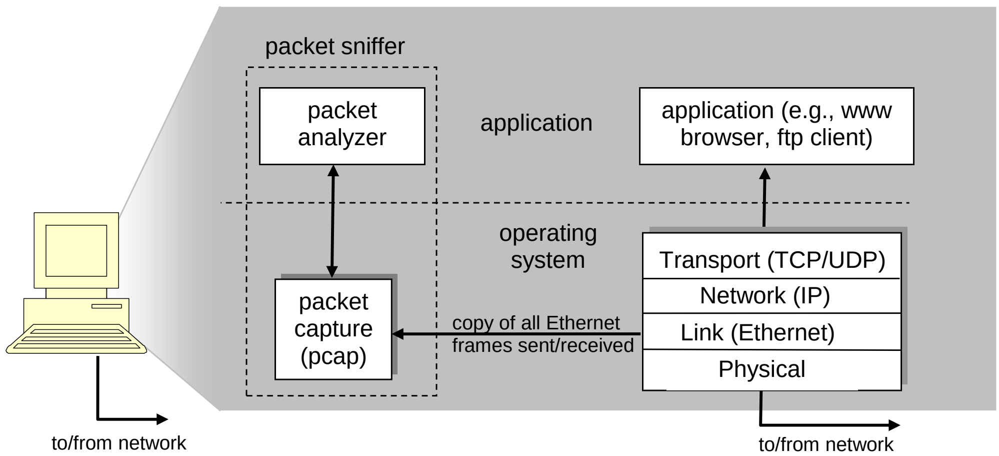


Figure A.1: Packet sniffer structure

The second component of a packet sniffer is the **packet analyzer**, which displays the contents of all fields within a protocol message. In order to do so, the packet analyzer must "understand" the structure of all messages exchanged by protocols. The packet analyzer understands the format of Ethernet frames, and so can identify the IP datagram within an Ethernet frame. It also understands the IP datagram format, so that it can extract the TCP segment within the IP datagram. Finally, it understands the TCP segment structure, so it can extract the HTTP message contained in the TCP segment. Finally, it understands the HTTP protocol and so, for example, knows that the first bytes of an HTTP message will contain the string "GET," "POST," or "HEAD".

2.0 Getting Wireshark Ready

- Download and install the Wireshark software
- Run Wireshark. Wireshark startup screen shown in Figure A.2.

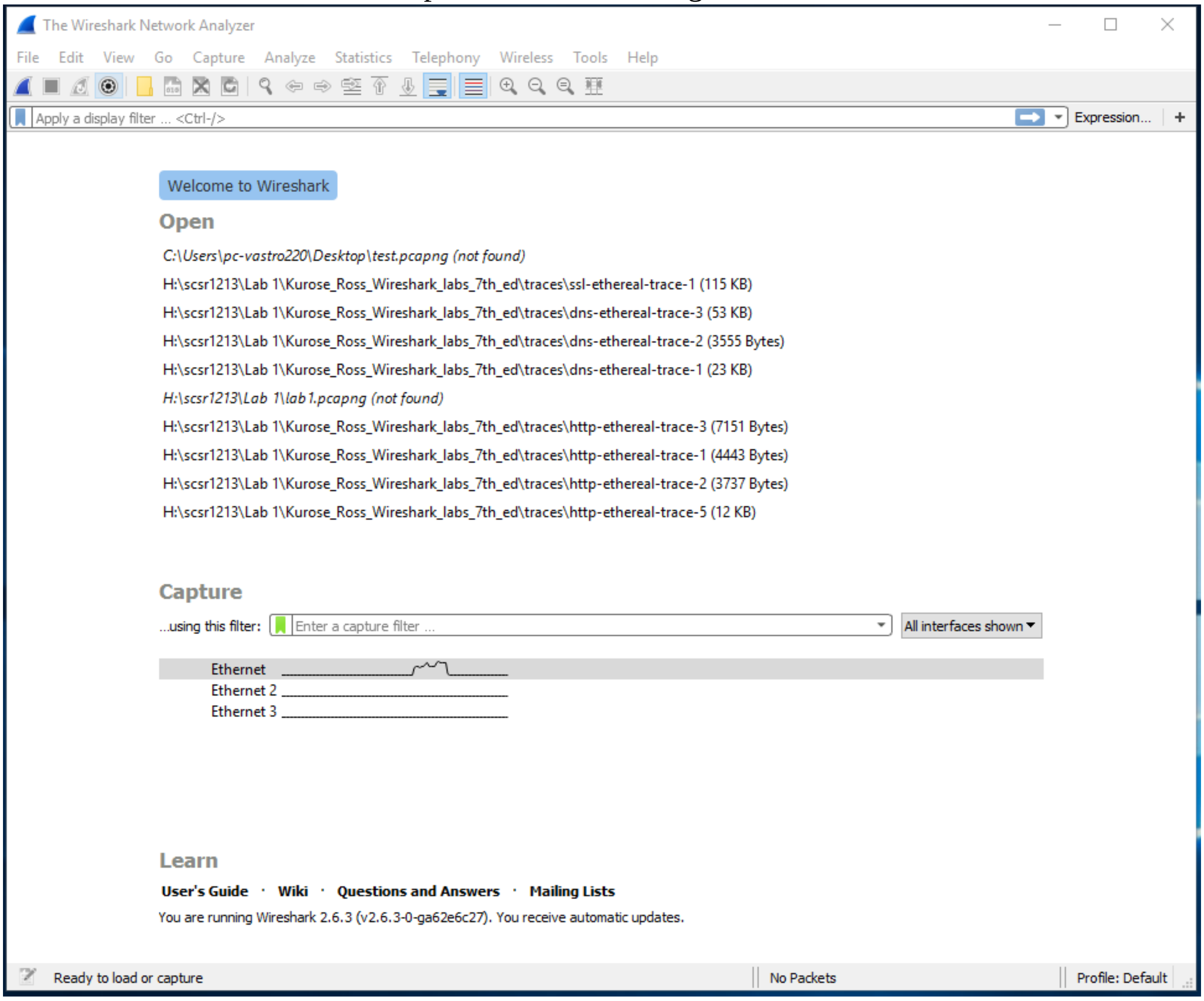


Figure A.2: Initial Wireshark startup screen

- The Wireshark interface has five major components as shown in Figure A.3.

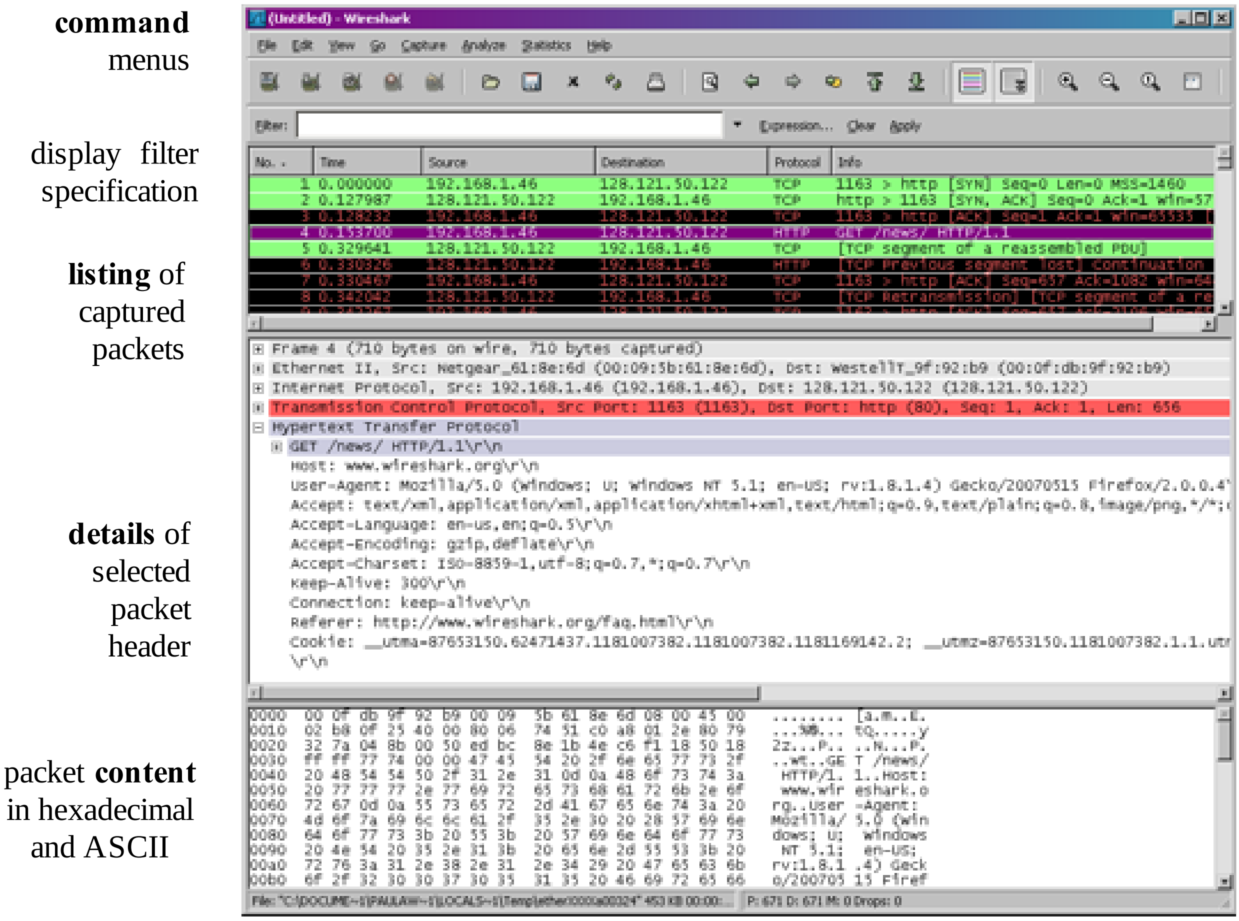


Figure A.3: Wireshark Graphical User Interface, during packet capture and

- The **command menus** are standard pulldown menus located at the top of the window.
- The **packet display filter field,** into which a protocol name or other information can be entered in order to filter the information displayed in the packet-listing window.
- The **packet-listing window** displays a one-line summary for each packet captured, including the packet number, the time at which the packet was captured, the packet's source and destination addresses, the protocol type, and protocol-specific information contained in the packet.
- The **packet-header details window** provides details about the packet selected (highlighted) in the packet-listing window. These details include information about the Ethernet frame and IP datagram that contains this packet. The amount of Ethernet and IP-layer detail displayed can be expanded or minimized by clicking on the plus minus boxes to the left of the Ethernet frame or IP datagram line in the packet details window. If the packet has been carried over TCP or UDP, TCP or UDP details will also be displayed, which can similarly be expanded or minimized. Finally, details about the highest-level protocol that sent or received this packet are also provided.
- The **packet-contents window** displays the entire contents of the captured frame, in both ASCII and hexadecimal format.

3.0 Test Run Wireshark

- Start up the Wireshark software.
- To begin packet capture, select the Capture pull down menu and pick Options menu. Select appropriate interfaces on your compute and click Start button to begin packet capture. Refer to Figure A.4

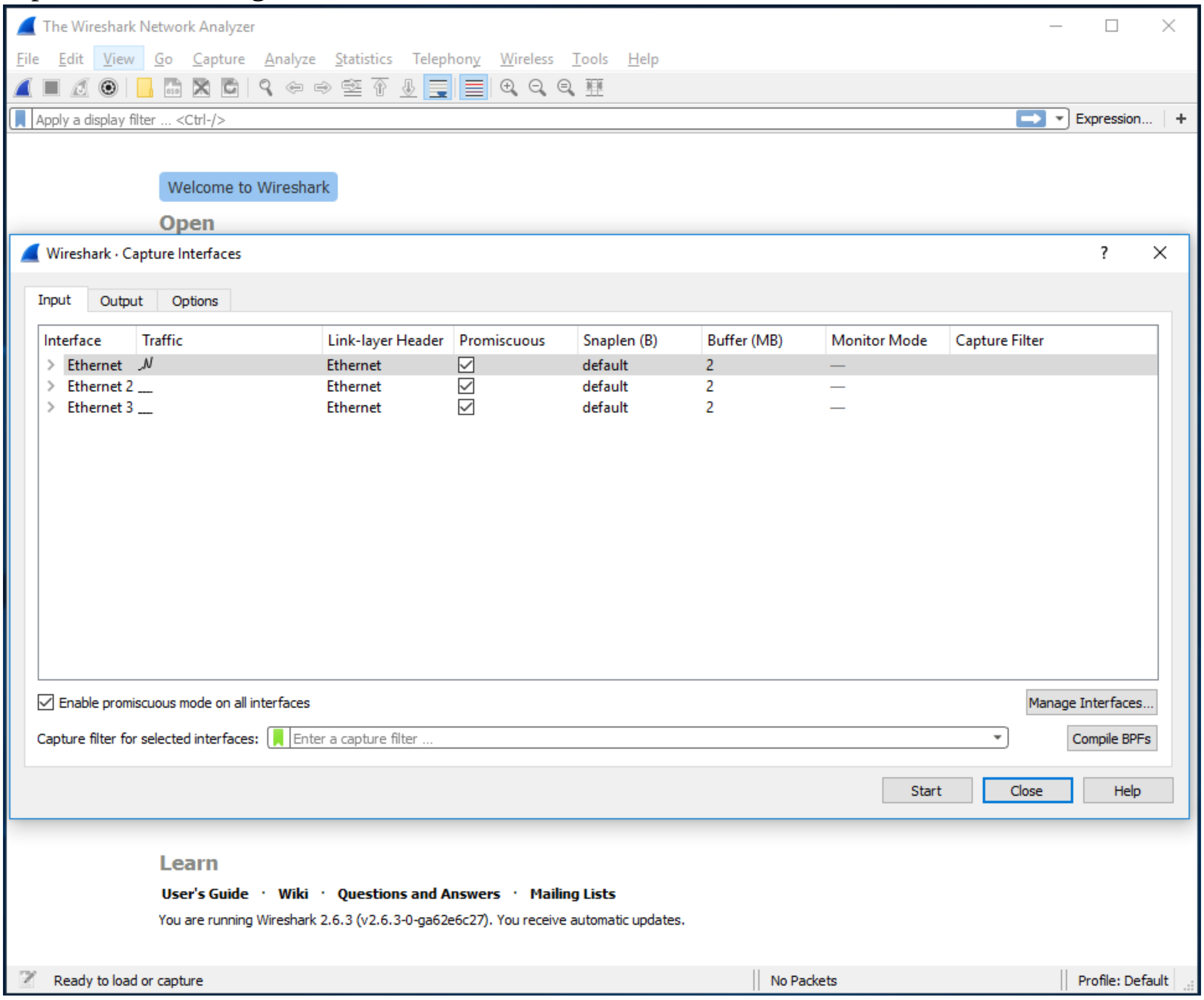


Figure A.4: Capture and Options Menu

- Once you begin packet capture, result will be shown as in Figure A.5.

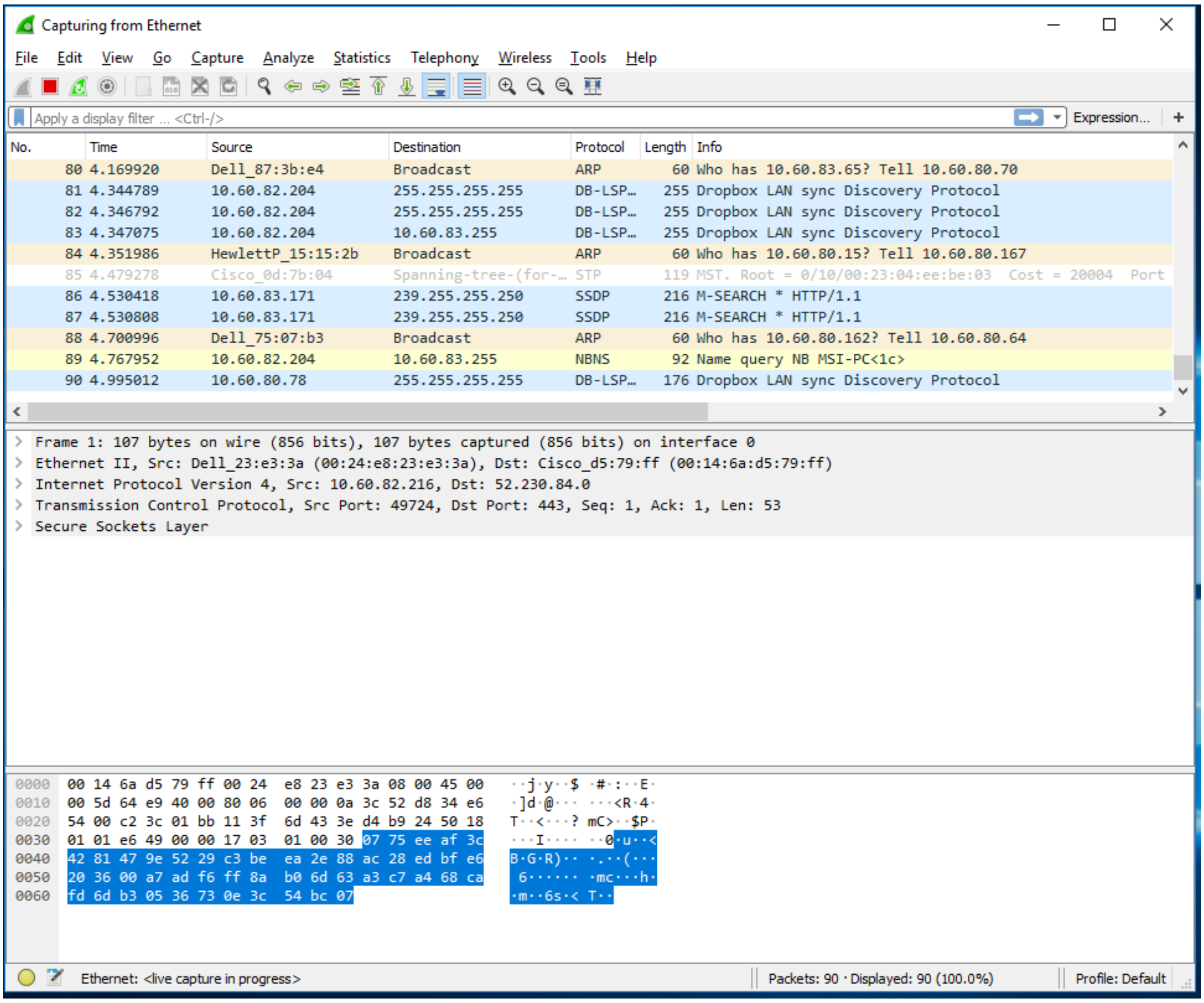


Figure A.5: Wireshark packet capture result

- By selecting Capture pulldown menu and selecting Stop, you can stop packet capture.

- Type “arp” in packet display filter field and press Enter key. This will cause only ARP message to be displayed in the packet-listing window as shown in Figure A.6.

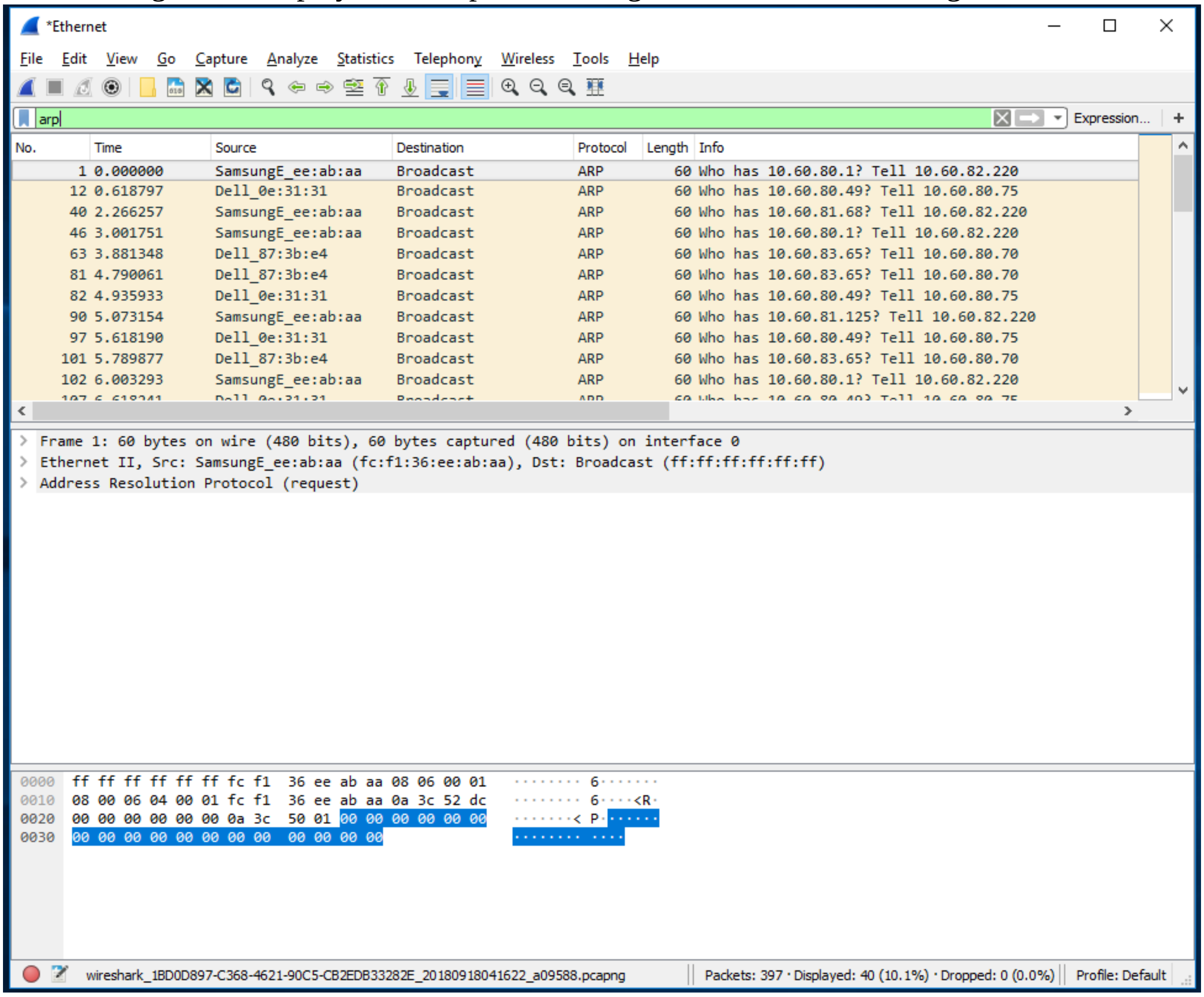


Figure A.6: ARP packet capture

- To save the trace result, use File pulldown menu and select Save function as shown in Figure A.7.

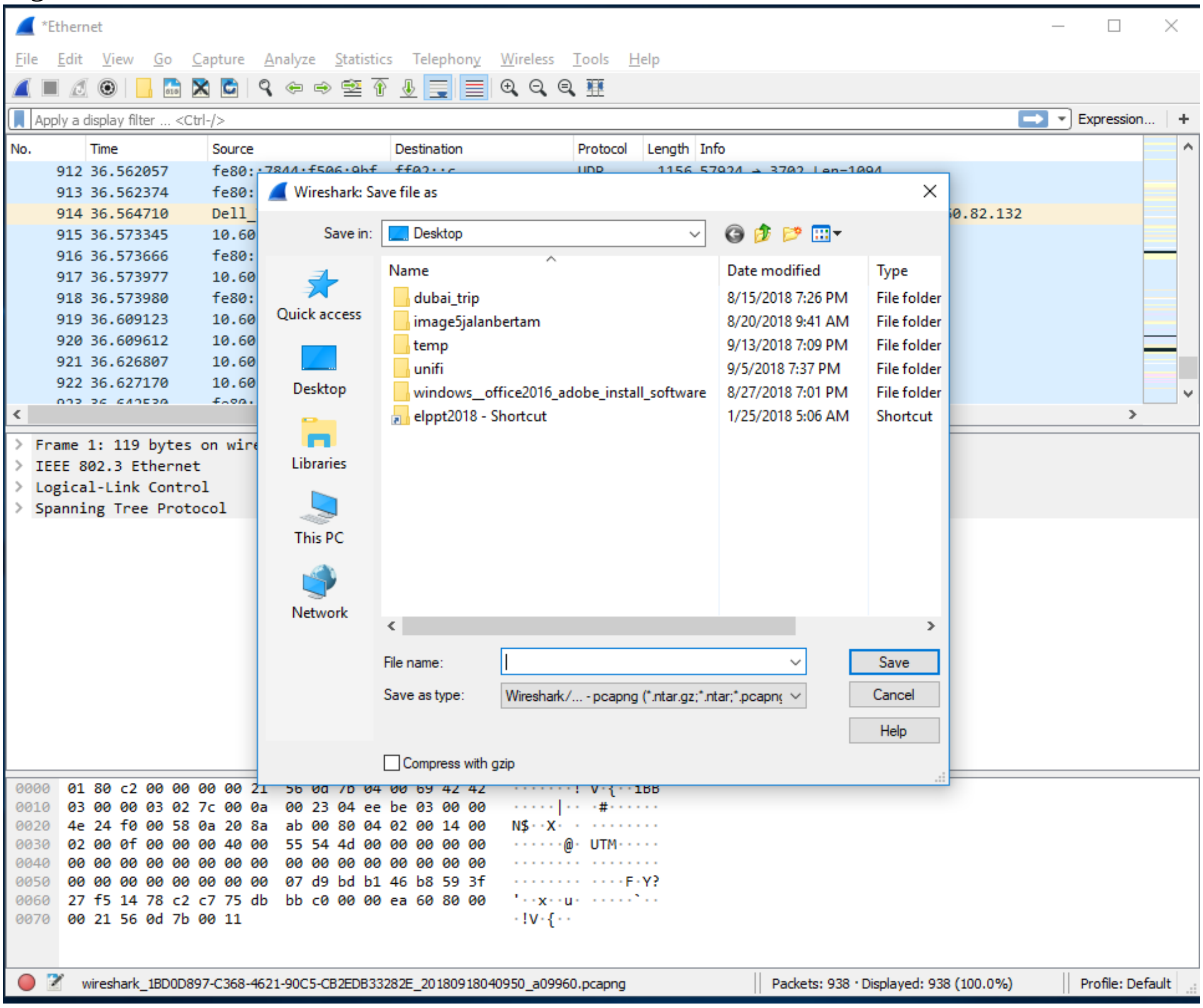


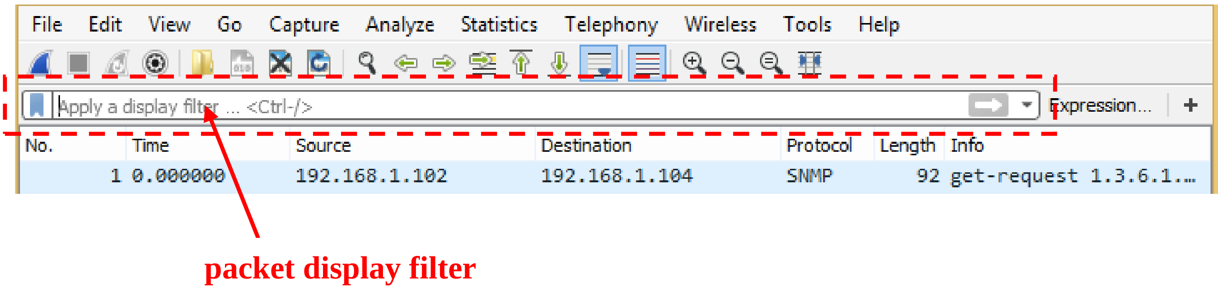
Figure A.7: Save Wireshark trace result

PART B: HTTP Trace

In this part, we'll explore several aspects of the HTTP protocol: the basic GET/response interaction, HTTP message formats and retrieving HTML files with embedded objects. Before beginning these labs, you might want to review Section 2.2 of the textbook.

B.1 The Basic HTTP GET/response interaction

- Open packet trace file **lab1-http-B01.pcapng**.
- Enter "**http**" (just the letters, not the quotation marks) in the **packet display filter field**, so that only captured HTTP messages will be displayed later in the packet-listing window. Refer to figure below:



- By looking at the information in the HTTP GET and response messages, answer the following questions:

1. What version of HTTP is the server running?
   The server is running HTTP Version 1.1, as indicated by the captured HTTP messages, which show that all requests and responses are using the HTTP/1.1 protocol as shown in Figure B.1.1 and Figure B.1.2.

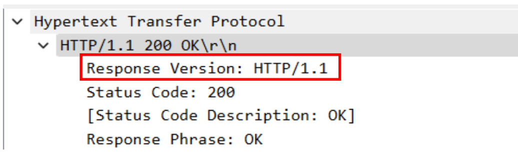
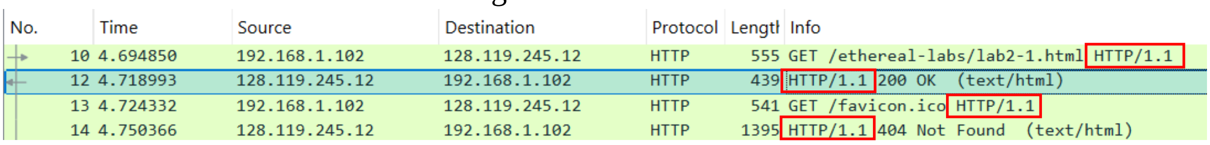


Figure B.1.1.1



No.	Time	Source	Destination	Protocol	Lengtl	Info
10	4.694850	192.168.1.102	128.119.245.12	HTTP	555	GET /ethereal-labs/lab2-1.html HTTP/1.1
12	4.718993	128.119.245.12	192.168.1.102	HTTP	439	HTTP/1.1 200 OK (text/html)
13	4.724332	192.168.1.102	128.119.245.12	HTTP	541	GET /favicon.ico HTTP/1.1
14	4.750366	128.119.245.12	192.168.1.102	HTTP	1395	HTTP/1.1 404 Not Found (text/html)

Figure B.1.1.2

2. What is the IP address of the client computer?
   The IP address of the client computer is 192.168.1.102, which is the destination address.

```
v Internet Protocol Version 4, Src: 128.119.245.12, Dst: 192.168.1.102
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 425
    Identification: 0xb6fa (46842)
  > 010. .... = Flags: 0x2, Don't fragment
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 55
    Protocol: TCP (6)
    Header Checksum: 0x53c2 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 128.119.245.12
    Destination Address: 192.168.1.102
    [Stream index: 2]
```

Figure B.1.2

3. What is the IP address of the gaia.cs.umass.edu server?
   The IP address of the client computer is 128.119.245.12, which is the source address.

```
v Internet Protocol Version 4, Src: 128.119.245.12, Dst: 192.168.1.102
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 425
    Identification: 0xb6fa (46842)
  > 010. .... = Flags: 0x2, Don't fragment
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 55
    Protocol: TCP (6)
    Header Checksum: 0x53c2 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 128.119.245.12
    Destination Address: 192.168.1.102
    [Stream index: 2]
```

Figure B.1.3

4. How many bytes of content are being returned to client browser?
   73 bytes

```
v Hypertext Transfer Protocol
  > HTTP/1.1 200 OK\r\n
    Date: Tue, 23 Sep 2003 05:29:50 GMT\r\n
    Server: Apache/2.0.40 (Red Hat Linux)\r\n
    Last-Modified: Tue, 23 Sep 2003 05:29:00 GMT\r\n
    ETag: "1bfed-49-79d5bf00"\r\n
    Accept-Ranges: bytes\r\n
  v Content-Length: 73\r\n
       [Content length: 73]
    Keep-Alive: timeout=10, max=100\r\n
    Connection: Keep-Alive\r\n
    Content-Type: text/html; charset=ISO-8859-1\r\n
    \r\n
    [Request in frame: 10]
    [Time since request: 0.024143000 seconds]
    [Request URI: /ethereal-labs/lab2-1.html]
    [Full request URI: http://gaia.cs.umass.edu/ethereal-labs/lab2-1.html]
    File Data: 73 bytes
```

Figure B.1.4

5. What is the status code returned from the server to client browser?

200

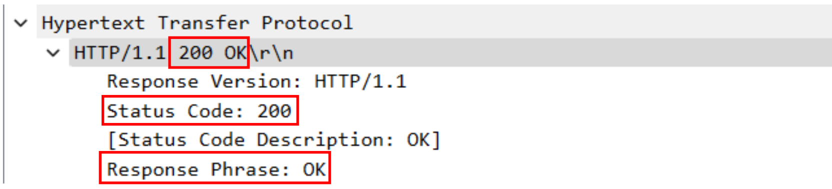


Figure B.1.5.1

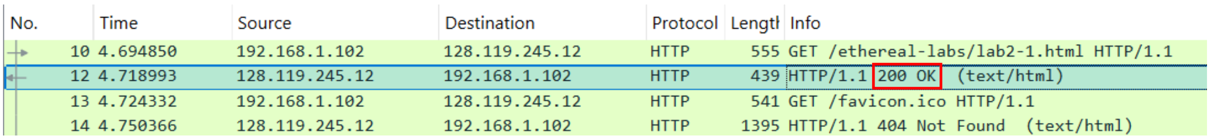


Figure B.1.5.2

B.2 The HTTP CONDITIONAL GET/response interaction

- Open packet trace file **lab1-http-B02.pcapng**.
- By looking at the information in the HTTP GET and response messages, answer the following questions:

1. Inspect the contents of the first HTTP GET request from your browser to the server. Do you see an "IF-MODIFIED-SINCE" line in the HTTP GET?

   No, there is no "IF-MODIFIED-SINCE" line in the HTTP GET.

```
v Hypertext Transfer Protocol
  v GET /ethereal-labs/lab2-2.html HTTP/1.1\r\n
      Request Method: GET
      Request URI: /ethereal-labs/lab2-2.html
      Request Version: HTTP/1.1
    Host: gaia.cs.umass.edu\r\n
```

Figure B.2.1

2. Inspect the contents of the server response after the first GET request from client. Did the server explicitly return the contents of the file? How can you tell?

   Yes, since the status code returned from the server to the client browser for the first GET request is 200 with the response phase "OK" which means that the file has successfully retrieved and returned. This confirms that the server explicitly returned the contents of the requested file.

```
v Hypertext Transfer Protocol
  v HTTP/1.1 200 OK\r\n
      Response Version: HTTP/1.1
      Status Code: 200
      [Status Code Description: OK]
      Response Phrase: OK
```

Figure B.2.2

3. Now inspect the contents of the second HTTP GET request from your browser to the server. Do you see an "IF-MODIFIED-SINCE:" line in the HTTP GET? If so, what information follows the "IF-MODIFIED-SINCE:" header?

   Yes, the IF-MODIFIED-SINCE line is present in the second HTTP GET request and the information follows the "If-Modified-Since:" header is "Tue, 23 Sep 2003 05:35:00 GMT\r\n".

```
Hypertext Transfer Protocol
  GET /ethereal-labs/lab2-2.html HTTP/1.1\r\n
      Request Method: GET
      Request URI: /ethereal-labs/lab2-2.html
      Request Version: HTTP/1.1
  Host: gaia.cs.umass.edu\r\n
  User-Agent: Mozilla/5.0 (Windows; U; Windows NT 5.1; en-US; rv:1.0.2) Gecko/20021120 Netscape/7.01\r\n
  Accept: text/xml,application/xml,application/xhtml+xml,text/html;q=0.9,text/plain;q=0.8,video/x-mng,image/png,
  Accept-Language: en-us, en;q=0.50\r\n
  Accept-Encoding: gzip, deflate, compress;q=0.9\r\n
  Accept-Charset: ISO-8859-1, utf-8;q=0.66, *;q=0.66\r\n
  Keep-Alive: 300\r\n
  Connection: keep-alive\r\n
  If-Modified-Since: Tue, 23 Sep 2003 05:35:00 GMT\r\n
  If-None-Match: "1bfef-173-8f4ae900"\r\n
  Cache-Control: max-age=0\r\n
  \r\n
  [Response in frame: 15]
  [Full request URI: http://gaia.cs.umass.edu/ethereal-labs/lab2-2.html]
```

Figure B.2.3

4. What is the HTTP status code and phrase returned from the server in response to this second HTTP GET? Did the server explicitly return the contents of the file? Explain.

   The HTTP status code returned by the server is 304, with the phrase "Not Modified". No, the server did not explicitly return the contents of the file because the 304 Not Modified status indicates that the requested source has not changed since the date specified in the IF-MODIFIED-SINCE header. This response allows the browser to use its cached version of the file, reducing unnecessary and redundancy data transfer and improving efficiency.

```
Hypertext Transfer Protocol
  HTTP/1.1 304 Not Modified\r\n
      Response Version: HTTP/1.1
      Status Code: 304
      [Status Code Description: Not Modified]
      Response Phrase: Not Modified
```

Figure B.2.4

B.3 HTML Documents with Embedded Objects

- Open packet trace file **lab1-http-B03.pcapng**.
- By looking at the information in the HTTP GET and response messages, answer the following questions:

1. How many HTTP GET request messages did client browser send?

   3

No.	Time	Source	Destination	Protocol	Lengtl	Info
10	7.236929	192.168.1.102	128.119.245.12	HTTP	555	GET /ethereal-labs/lab2-4.html HTTP/1.1
12	7.260813	128.119.245.12	192.168.1.102	HTTP	1057	HTTP/1.1 200 OK (text/html)
17	7.305485	192.168.1.102	165.193.123.218	HTTP	625	GET /catalog/images/pearson-logo-footer.gif HTTP/1.1
20	7.308803	192.168.1.102	134.241.6.82	HTTP	609	GET /~kurose/cover.jpg HTTP/1.1
25	7.333054	165.193.123.218	192.168.1.102	HTTP	912	HTTP/1.1 200 OK (GIF89a)
54	7.589877	134.241.6.82	192.168.1.102	HTTP	1096	HTTP/1.0 200 Document follows (JPEG JFIF image)

Figure B.3.1

2. To which Internet addresses were these GET requests sent?
   First GET requests: 128.119.245.12
   Second GET requests: 165.193.123.218
   Third GET requests: 134.241.6.82

No.	Time	Source	Destination	Protocol	Lengtl	Info
10	7.236929	192.168.1.102	128.119.245.12	HTTP	555	GET /ethereal-labs/lab2-4.html HTTP/1.1
12	7.260813	128.119.245.12	192.168.1.102	HTTP	1057	HTTP/1.1 200 OK (text/html)
17	7.305485	192.168.1.102	165.193.123.218	HTTP	625	GET /catalog/images/pearson-logo-footer.gif HTTP/1.1
20	7.308803	192.168.1.102	134.241.6.82	HTTP	609	GET /~kurose/cover.jpg HTTP/1.1
25	7.333054	165.193.123.218	192.168.1.102	HTTP	912	HTTP/1.1 200 OK (GIF89a)
54	7.589877	134.241.6.82	192.168.1.102	HTTP	1096	HTTP/1.0 200 Document follows (JPEG JFIF image)

Figure B.3.2

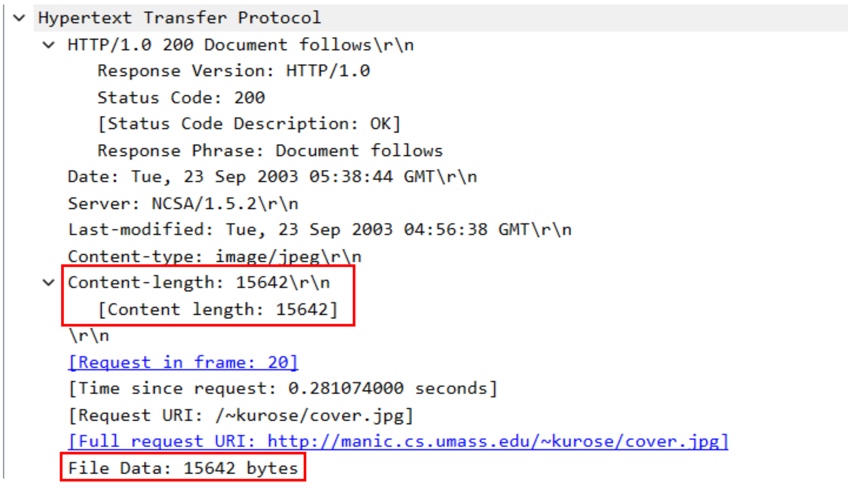
3. How many bytes of content are being returned to client browser for the **pearson-logo-footer.gif** image file?
   The bytes of the pearson-logo-footer.gif image file returned to client browser is 3357 bytes.

```
Hypertext Transfer Protocol
  HTTP/1.1 200 OK\r\n
    Response Version: HTTP/1.1
    Status Code: 200
    [Status Code Description: OK]
    Response Phrase: OK
  Server: Netscape-Enterprise/3.6 SP3\r\n
  Date: Sun, 21 Sep 2003 06:00:35 GMT\r\n
  Content-type: image/gif\r\n
  Etag: "6fc149-d1d-3ef0b3f8"\r\n
  Last-modified: Wed, 18 Jun 2003 18:48:24 GMT\r\n
  Content-length: 3357\r\n
  Accept-ranges: bytes\r\n
  Connection: keep-alive\r\n
  \r\n
  [Request in frame: 17]
  [Time since request: 0.027569000 seconds]
  [Request URI: /catalog/images/pearson-logo-footer.gif]
  [Full request URI: http://www.aw-bc.com/catalog/images/pearson-logo-footer.gif]
  File Data: 3357 bytes
```

Figure B.3.3

4. How many bytes of content are being returned to client browser for the **cover.jpg** image file?

The bytes of cover.jpg image file returned to client browser is 15642 bytes.



```
Hypertext Transfer Protocol
  HTTP/1.0 200 Document follows\r\n
    Response Version: HTTP/1.0
    Status Code: 200
    [Status Code Description: OK]
    Response Phrase: Document follows
  Date: Tue, 23 Sep 2003 05:38:44 GMT\r\n
  Server: NCSA/1.5.2\r\n
  Last-modified: Tue, 23 Sep 2003 04:56:38 GMT\r\n
  Content-type: image/jpeg\r\n
  Content-length: 15642\r\n
    [Content length: 15642]
  \r\n
  [Request in frame: 20]
  [Time since request: 0.281074000 seconds]
  [Request URI: /~kurose/cover.jpg]
  [Full request URI: http://manic.cs.umass.edu/~kurose/cover.jpg]
  File Data: 15642 bytes
```

Figure B.2.4

PART C: DNS Trace

1.0 nslookup

nslookup tool allows the host running the tool to query any specified DNS server for a DNS record. The queried DNS server can be a root DNS server, a top-level-domain DNS server, an authoritative DNS server, or an intermediate DNS server. To accomplish this task, nslookup sends a DNS query to the specified DNS server, receives a DNS reply from that same DNS server, and displays the result.

- To run it in Windows, open the Command Prompt (cmd) and run nslookup on the command line as shown in Figure C.1 and Figure C.2

```
C:\WINDOWS\system32\cmd.exe

C:\Users\pc-vastro220>nslookup yahoo.com
Server:  ns1.utm.my
Address:  161.139.250.2

Non-authoritative answer:
Name:    yahoo.com
Addresses:  2001:4998:58:1836::11
          2001:4998:c:1023::4
          2001:4998:44:41d::4
          2001:4998:44:41d::3
          2001:4998:58:1836::10
          2001:4998:c:1023::5
          72.30.35.10
          98.137.246.7
          98.137.246.8
          98.138.219.231
          98.138.219.232
          72.30.35.9


C:\Users\pc-vastro220>nslookup -type=any yahoo.com
Server:  jb.utm.my
Address:  161.139.250.2

Non-authoritative answer:
yahoo.com       MX preference = 1, mail exchanger = mta5.am0.yahoodns.net
yahoo.com       MX preference = 1, mail exchanger = mta6.am0.yahoodns.net
yahoo.com       MX preference = 1, mail exchanger = mta7.am0.yahoodns.net
yahoo.com       AAAA IPv6 address = 2001:4998:c:1023::4
yahoo.com       AAAA IPv6 address = 2001:4998:44:41d::4
yahoo.com       AAAA IPv6 address = 2001:4998:44:41d::3
yahoo.com       AAAA IPv6 address = 2001:4998:58:1836::10
yahoo.com       AAAA IPv6 address = 2001:4998:c:1023::5
yahoo.com       AAAA IPv6 address = 2001:4998:58:1836::11
yahoo.com       nameserver = ns5.yahoo.com
yahoo.com       nameserver = ns2.yahoo.com
yahoo.com       nameserver = ns4.yahoo.com
yahoo.com       nameserver = ns1.yahoo.com
yahoo.com       nameserver = ns3.yahoo.com
yahoo.com       internet address = 98.137.246.7
yahoo.com       internet address = 98.137.246.8
yahoo.com       internet address = 98.138.219.231
yahoo.com       internet address = 98.138.219.232
yahoo.com       internet address = 72.30.35.9
yahoo.com       internet address = 72.30.35.10

C:\Users\pc-vastro220>
```

Figure C.1: nslookup result

```
C:\WINDOWS\system32\cmd.exe

C:\Users\pc-vastro220>nslookup google.com ns1.time.net.my
Server:  ns1.test.time.net.my
Address:  203.121.16.85

Non-authoritative answer:
Name:    google.com
Addresses:  2404:6800:4001:804::200e
          172.217.24.174


C:\Users\pc-vastro220>
```

Figure C.2: nslookup result

1. Run nslookup to obtain the IP address of a www.microsoft.com server. What is the IP address of that server? Add screenshot to your answer.
   184.25.221.172

```
C:\Users\Stell>nslookup www.microsoft.com
Server:  ns1.utm.my
Address:  161.139.250.2

Non-authoritative answer:
Name:    e13678.dscb.akamaiedge.net
Addresses:  2600:1413:a000:794::356e
          2600:1413:a000:7b9::356e
          2600:1413:a000:7a4::356e
          2600:1413:a000:7aa::356e
          2600:1413:a000:7ad::356e
          184.25.221.172
Aliases:  www.microsoft.com
          www.microsoft.com-c-3.edgekey.net
          www.microsoft.com-c-3.edgekey.net.globalredir.akadns.net
```

Figure C.1.1

2. Run nslookup to determine the non-authoritative DNS servers for domain microsoft.com. Add screenshot to your answer.
   ns1-39.azure-dns.com
   ns4-39.azure-dns.info
   ns3-39.azure-dns.org
   ns2-39.azure-dns.net

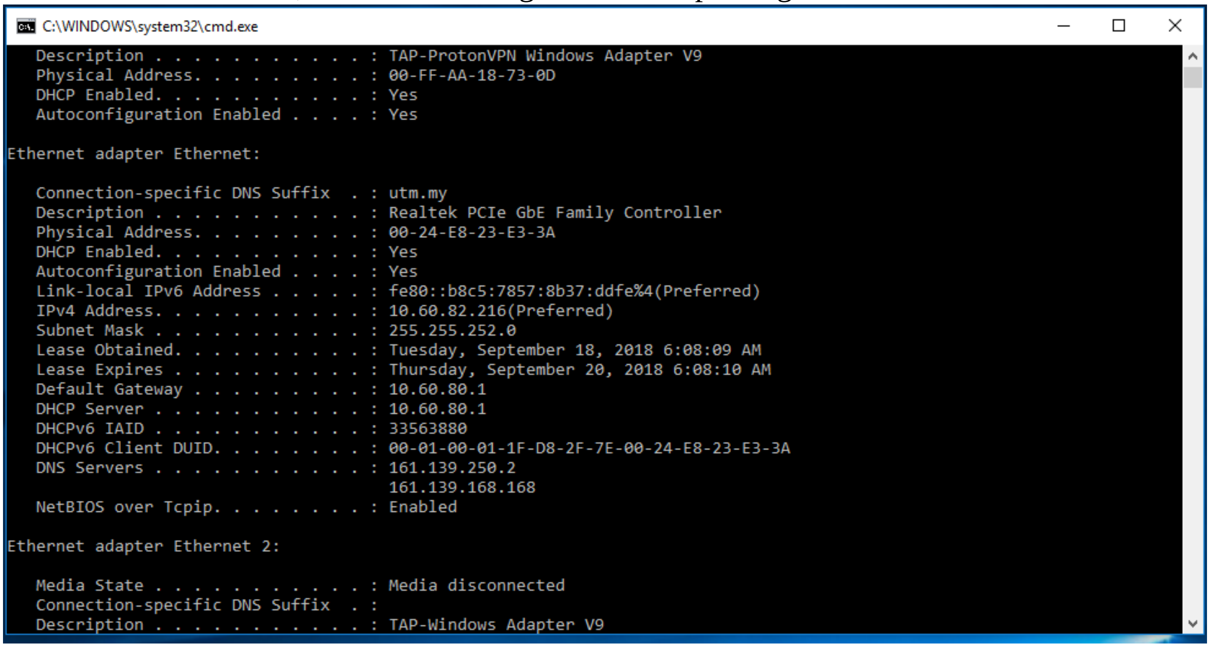
```
C:\Users\Stell>nslookup -type=ns microsoft.com
Server:  ns1.utm.my
Address:  161.139.250.2

Non-authoritative answer:
microsoft.com   nameserver = ns1-39.azure-dns.com
microsoft.com   nameserver = ns4-39.azure-dns.info
microsoft.com   nameserver = ns3-39.azure-dns.org
microsoft.com   nameserver = ns2-39.azure-dns.net

ns1-39.azure-dns.com    internet address = 150.171.10.39
ns2-39.azure-dns.net    internet address = 150.171.16.39
ns3-39.azure-dns.org    internet address = 13.107.222.39
ns4-39.azure-dns.info   internet address = 13.107.206.39
```

Figure C.1.2

2.0 ipconfig

ipconfig can be used to show your current TCP/IP information, including your address, DNS server addresses, adapter type and so on.

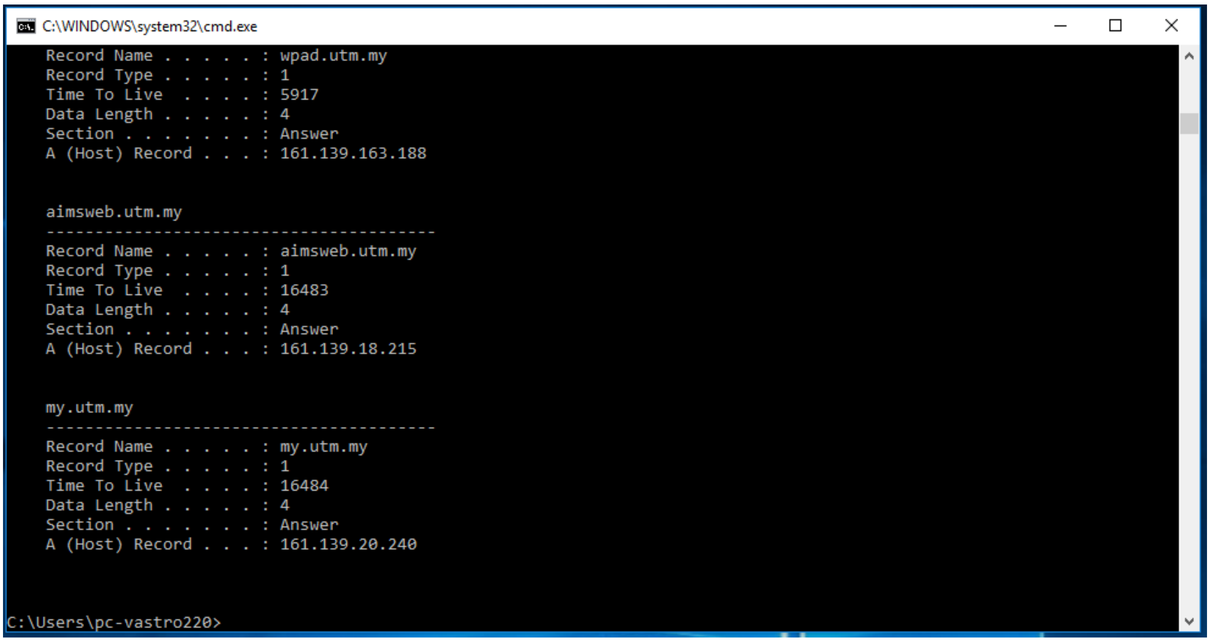
- Information about host, use the following command: ipconfig /all



```
C:\WINDOWS\system32\cmd.exe
   Description . . . . . . . . . . . : TAP-ProtonVPN Windows Adapter V9
   Physical Address. . . . . . . . . : 00-FF-AA-18-73-0D
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . : utm.my
   Description . . . . . . . . . . . : Realtek PCIe GbE Family Controller
   Physical Address. . . . . . . . . : 00-24-E8-23-E3-3A
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::b8c5:7857:8b37:ddfe%4(Preferred)
   IPv4 Address. . . . . . . . . . . : 10.60.82.216(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.252.0
   Lease Obtained. . . . . . . . . . : Tuesday, September 18, 2018 6:08:09 AM
   Lease Expires . . . . . . . . . . : Thursday, September 20, 2018 6:08:10 AM
   Default Gateway . . . . . . . . . : 10.60.80.1
   DHCP Server . . . . . . . . . . . : 10.60.80.1
   DHCPv6 IAID . . . . . . . . . . . : 33563880
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-1F-D8-2F-7E-00-24-E8-23-E3-3A
   DNS Servers . . . . . . . . . . . : 161.139.250.2
                                       161.139.168.168
   NetBIOS over Tcpip. . . . . . . . : Enabled

Ethernet adapter Ethernet 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : TAP-Windows Adapter V9
```

Figure C.3: ipconfig /all result

- ipconfig is also very useful for managing the DNS information stored in your host. Each entry shows the remaining Time to Live (TTL) in seconds.
  Command: ipconfig /displaydns



```
C:\WINDOWS\system32\cmd.exe
    Record Name . . . . . : wpad.utm.my
    Record Type . . . . . : 1
    Time To Live  . . . . : 5917
    Data Length . . . . . : 4
    Section . . . . . . . : Answer
    A (Host) Record . . . : 161.139.163.188


    aimsweb.utm.my
    ----------------------------------------
    Record Name . . . . . : aimsweb.utm.my
    Record Type . . . . . : 1
    Time To Live  . . . . : 16483
    Data Length . . . . . : 4
    Section . . . . . . . : Answer
    A (Host) Record . . . : 161.139.18.215


    my.utm.my
    ----------------------------------------
    Record Name . . . . . : my.utm.my
    Record Type . . . . . : 1
    Time To Live  . . . . : 16484
    Data Length . . . . . : 4
    Section . . . . . . . : Answer
    A (Host) Record . . . : 161.139.20.240


C:\Users\pc-vastro220>
```

Figure C.4: ipconfig /displaydns result

- Flushing the DNS cache clears all entries and reloads the entries from the hosts file.

Command: ipconfig /flushdns

Figure C.5: ipconfig /flushdns result

3.0 Tracing DNS with Wireshark

- Open packet trace file dns-trace-1. Answer the following questions.

1. Locate the DNS query and response messages. Are then sent over UDP or TCP? Add screenshots in your answer.
   Both send over UDP

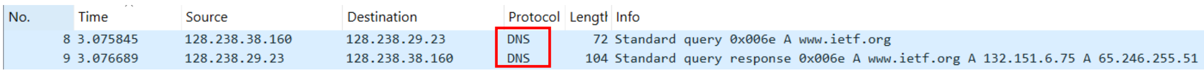


Figure C.3.1.1

DNS query message:

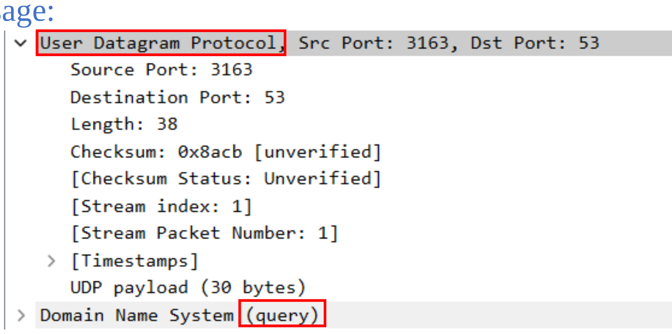


Figure C.3.1.2

DNS response message:

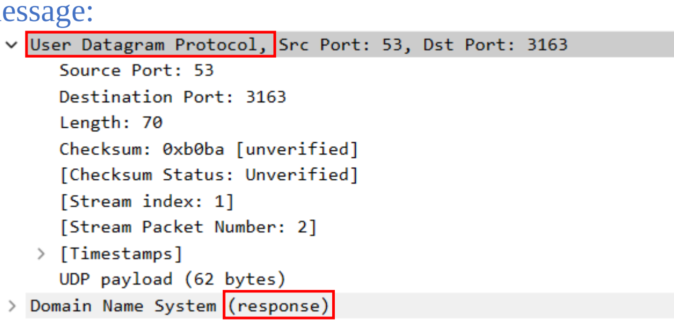


Figure C.3.1.3

2. What is the destination port for the DNS query message? What is the source port of DNS response message? Add screenshots in your answer.

   Destination port for DNS query message: 53

```
v User Datagram Protocol, Src Port: 3163, Dst Port: 53
    Source Port: 3163
    Destination Port: 53
    Length: 38
    Checksum: 0x8acb [unverified]
    [Checksum Status: Unverified]
    [Stream index: 1]
    [Stream Packet Number: 1]
  > [Timestamps]
    UDP payload (30 bytes)
> Domain Name System (query)
```

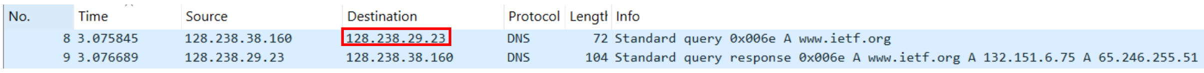
Figure C.3.2.1

   Source port for DNS response message: 53

```
v User Datagram Protocol, Src Port: 53, Dst Port: 3163
    Source Port: 53
    Destination Port: 3163
    Length: 70
    Checksum: 0xb0ba [unverified]
    [Checksum Status: Unverified]
    [Stream index: 1]
    [Stream Packet Number: 2]
  > [Timestamps]
    UDP payload (62 bytes)
> Domain Name System (response)
```

Figure C.3.2.2

3. To what IP address is the DNS query message sent? Add screenshots in your answer.
   128.238.29.23



No.	Time	Source	Destination	Protocol	Length	Info
8	3.075845	128.238.38.160	128.238.29.23	DNS	72	Standard query 0x006e A www.ietf.org
9	3.076689	128.238.29.23	128.238.38.160	DNS	104	Standard query response 0x006e A www.ietf.org A 132.151.6.75 A 65.246.255.51

Figure C.3.3

4. Examine the DNS query message. What "Type" of DNS query is it? Does the query message contain any "answers"? Add screenshots in your answer.
   The Type" of DNS query is Type A. The query message does not contain any "answers".

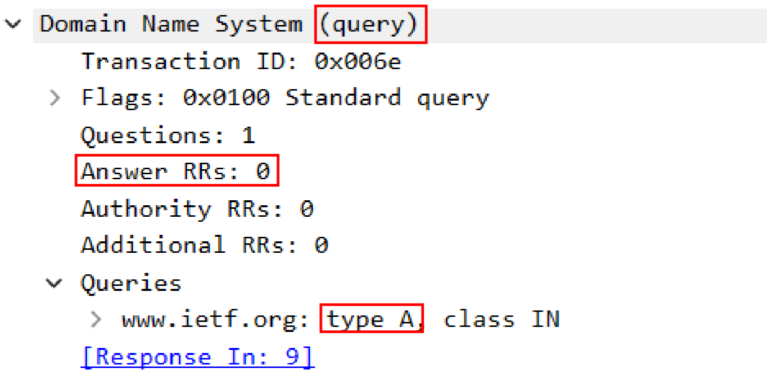


Figure C.3.4

5. Examine the DNS response message. How many “answers” are provided? What do each of these answers contain? Add screenshots in your answer.

   There are 2 “answers” provided. Each of these answers contain domain name, address type, class, time to live, data length and IP address.

```
v Domain Name System (response)
    Transaction ID: 0x006e
  > Flags: 0x8180 Standard query response, No error
    Questions: 1
    Answer RRs: 2
    Authority RRs: 0
    Additional RRs: 0
  v Queries
    > www.ietf.org: type A, class IN
```

Figure C.3.5.1

```
v Answers
  v www.ietf.org: type A, class IN, addr 132.151.6.75
      Name: www.ietf.org
      Type: A (1) (Host Address)
      Class: IN (0x0001)
      Time to live: 1678 (27 minutes, 58 seconds)
      Data length: 4
      Address: 132.151.6.75
  v www.ietf.org: type A, class IN, addr 65.246.255.51
      Name: www.ietf.org
      Type: A (1) (Host Address)
      Class: IN (0x0001)
      Time to live: 1678 (27 minutes, 58 seconds)
      Data length: 4
      Address: 65.246.255.51
  [Request In: 8]
  [Time: 0.000844000 seconds]
```

Figure C.3.5.2

6. Consider the subsequent TCP SYN packet sent by your host. Does the destination IP address of the SYN packet correspond to any of the IP addresses provided in the DNS response message? Add screenshots in your answer.
   Yes. The destination IP address of the SYN packet is correspond to the IP address provided in the DNS response message which is 132.151.6.75.

No.	Time	Source	Destination	Protocol	Length	Info
10	3.078479	128.238.38.160	132.151.6.75	TCP	62	3369 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM
11	3.096413	132.151.6.75	128.238.38.160	TCP	62	80 → 3369 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MSS=1380 SACK_PERM
12	3.096463	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=1 Ack=1 Win=64860 Len=0
13	3.096708	128.238.38.160	132.151.6.75	HTTP	429	GET / HTTP/1.1

Figure 3.6.1

```
Domain Name System (response)
    Transaction ID: 0x006e
  > Flags: 0x8180 Standard query response, No error
    Questions: 1
    Answer RRs: 2
    Authority RRs: 0
    Additional RRs: 0
  Queries
    > www.ietf.org: type A, class IN
```

Figure C.3.6.2

```
Answers
  www.ietf.org: type A, class IN, addr 132.151.6.75
      Name: www.ietf.org
      Type: A (1) (Host Address)
      Class: IN (0x0001)
      Time to live: 1678 (27 minutes, 58 seconds)
      Data length: 4
      Address: 132.151.6.75
  www.ietf.org: type A, class IN, addr 65.246.255.51
      Name: www.ietf.org
      Type: A (1) (Host Address)
      Class: IN (0x0001)
      Time to live: 1678 (27 minutes, 58 seconds)
      Data length: 4
      Address: 65.246.255.51
[Request In: 8]
[Time: 0.000844000 seconds]
```

Figure C.3.6.3

7. This web page contains images. Before retrieving each image, does your host issue new DNS queries?
   No. Line 28 is the line retrieve image and before that there is no host issue new DNS queries from line 10 until line 27.

No.	Time	Source	Destination	Protocol	Lengtl	Info
10	3.078479	128.238.38.160	132.151.6.75	TCP	62	3369 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM
11	3.096413	132.151.6.75	128.238.38.160	TCP	62	80 → 3369 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MSS=1380 SACK_PERM
12	3.096463	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=1 Ack=1 Win=64860 Len=0
13	3.096708	128.238.38.160	132.151.6.75	HTTP	429	GET / HTTP/1.1
14	3.111678	132.151.6.75	128.238.38.160	TCP	60	80 → 3369 [ACK] Seq=1 Ack=376 Win=6432 Len=0
15	3.120640	132.151.6.75	128.238.38.160	TCP	1434	80 → 3369 [ACK] Seq=1 Ack=376 Win=6432 Len=1380 [TCP PDU reassembled in 20]
16	3.128093	132.151.6.75	128.238.38.160	TCP	1434	80 → 3369 [ACK] Seq=1381 Ack=376 Win=6432 Len=1380 [TCP PDU reassembled in 20]
17	3.128148	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=376 Ack=2761 Win=64860 Len=0
18	3.148016	132.151.6.75	128.238.38.160	TCP	1434	80 → 3369 [ACK] Seq=2761 Ack=376 Win=6432 Len=1380 [TCP PDU reassembled in 20]
19	3.148069	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=376 Ack=4141 Win=64860 Len=0
20	3.153211	132.151.6.75	128.238.38.160	HTTP	1055	HTTP/1.1 200 OK (text/html)
21	3.153293	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=376 Ack=5143 Win=63859 Len=0
22	3.161867	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [FIN, ACK] Seq=376 Ack=5143 Win=63859 Len=0
23	3.174716	132.151.6.75	128.238.38.160	TCP	60	80 → 3369 [ACK] Seq=5143 Ack=377 Win=6432 Len=0
24	3.178159	128.238.38.160	132.151.6.75	TCP	62	3370 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM
25	3.179283	128.238.38.160	132.151.6.75	TCP	62	3371 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM
26	3.191649	132.151.6.75	128.238.38.160	TCP	62	80 → 3370 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MSS=1380 SACK_PERM
27	3.191726	128.238.38.160	132.151.6.75	TCP	54	3370 → 80 [ACK] Seq=1 Ack=1 Win=64860 Len=0
28	3.191998	128.238.38.160	132.151.6.75	HTTP	320	GET /images/ietflogo2e.gif HTTP/1.1

Figure C.3.7

- Open packet trace file dns-trace-2 for nslookup.
- We see from Wireshark that nslookup actually sent three DNS queries and received three DNS responses. For the purpose of this lab, ignore the first two sets of queries/responses, as they are specific to nslookup and are not normally generated by standard Internet applications. You should instead focus on the last query and response messages.
- Answer the following questions.

No.	Time	Source	Destination	Protocol	Lengtl	Info
15	4.951232	128.238.38.160	128.238.29.22	DNS	86	Standard query 0x0001 PTR 22.29.238.128.in-addr.arpa
16	4.951638	128.238.29.22	128.238.38.160	DNS	118	Standard query response 0x0001 PTR 22.29.238.128.in-addr.arpa PTR dns-prime.poly.edu
17	4.952571	128.238.38.160	128.238.29.22	DNS	80	Standard query 0x0002 A www.mit.edu.poly.edu
18	4.952953	128.238.29.22	128.238.38.160	DNS	139	Standard query response 0x0002 No such name A www.mit.edu.poly.edu SOA dns-prime.poly.edu
19	4.953172	128.238.38.160	128.238.29.22	DNS	71	Standard query 0x0003 A www.mit.edu
20	4.969929	128.238.29.22	128.238.38.160	DNS	196	Standard query response 0x0003 A www.mit.edu A 18.7.22.83 NS BITSY.mit.edu NS STRAWB.mit.e

8. What is the destination port for the DNS query message? What is the source port of DNS response message? Add screenshots in your answer.

Destination port for the DNS query message: 53

```
User Datagram Protocol, Src Port: 3742, Dst Port: 53
    Source Port: 3742
    Destination Port: 53
    Length: 37
    Checksum: 0x5890 [unverified]
    [Checksum Status: Unverified]
    [Stream index: 3]
    [Stream Packet Number: 1]
  > [Timestamps]
    UDP payload (29 bytes)
> Domain Name System (query)
```

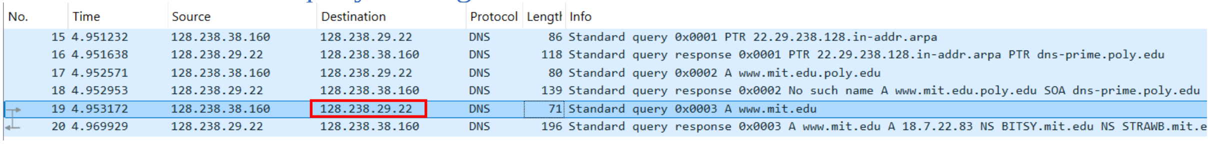
Figure C.3.8.1

Source port of DNS response message: 53

```
User Datagram Protocol, Src Port: 53, Dst Port: 3742
    Source Port: 53
    Destination Port: 3742
    Length: 162
    Checksum: 0xa318 [unverified]
    [Checksum Status: Unverified]
    [Stream index: 3]
    [Stream Packet Number: 2]
  > [Timestamps]
    UDP payload (154 bytes)
> Domain Name System (response)
```

Figure C.3.8.2

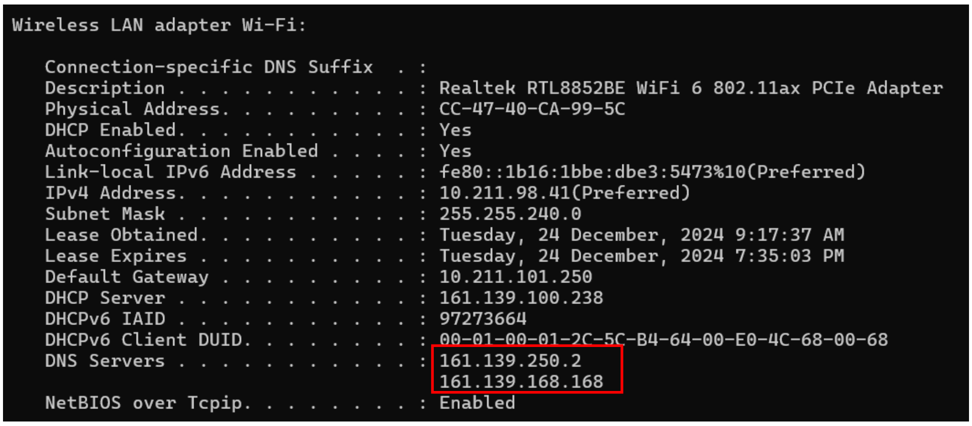
9. To what IP address is the DNS query message sent? Is this the IP address of your default local DNS server? Add screenshots in your answer.
IP address of the DNS query message sent: 128.238.29.22



No.	Time	Source	Destination	Protocol	Length	Info
15	4.951232	128.238.38.160	128.238.29.22	DNS	86	Standard query 0x0001 PTR 22.29.238.128.in-addr.arpa
16	4.951638	128.238.29.22	128.238.38.160	DNS	118	Standard query response 0x0001 PTR 22.29.238.128.in-addr.arpa PTR dns-prime.poly.edu
17	4.952571	128.238.38.160	128.238.29.22	DNS	80	Standard query 0x0002 A www.mit.edu.poly.edu
18	4.952953	128.238.29.22	128.238.38.160	DNS	139	Standard query response 0x0002 No such name A www.mit.edu.poly.edu SOA dns-prime.poly.edu
19	4.953172	128.238.38.160	128.238.29.22	DNS	71	Standard query 0x0003 A www.mit.edu
20	4.969929	128.238.29.22	128.238.38.160	DNS	196	Standard query response 0x0003 A www.mit.edu A 18.7.22.83 NS BITSY.mit.edu NS STRAWB.mit.e

Figure C.3.9.1

No. IP address of my default local DNS server: 161.139.250.2
161.139.168.168



```
Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Realtek RTL8852BE WiFi 6 802.11ax PCIe Adapter
   Physical Address. . . . . . . . . : CC-47-40-CA-99-5C
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::1b16:1bbe:dbe3:5473%10(Preferred)
   IPv4 Address. . . . . . . . . . . : 10.211.98.41(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.240.0
   Lease Obtained. . . . . . . . . . : Tuesday, 24 December, 2024 9:17:37 AM
   Lease Expires . . . . . . . . . . : Tuesday, 24 December, 2024 7:35:03 PM
   Default Gateway . . . . . . . . . : 10.211.101.250
   DHCP Server . . . . . . . . . . . : 161.139.100.238
   DHCPv6 IAID . . . . . . . . . . . : 97273664
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-2C-5C-B4-64-00-E0-4C-68-00-68
   DNS Servers . . . . . . . . . . . : 161.139.250.2
                                       161.139.168.168
   NetBIOS over Tcpip. . . . . . . . : Enabled
```

Figure C.3.9.2

10. Examine the DNS query message. What "Type" of DNS query is it? Does the query message contain any "answers"? Add screenshots in your answer.
The "Type" of DNS query is type A and the query message does not contain any "answer".

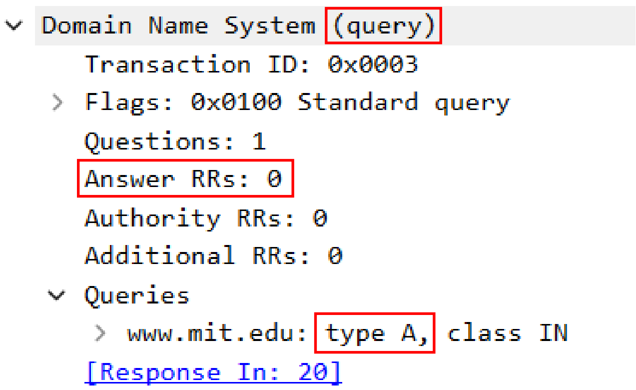
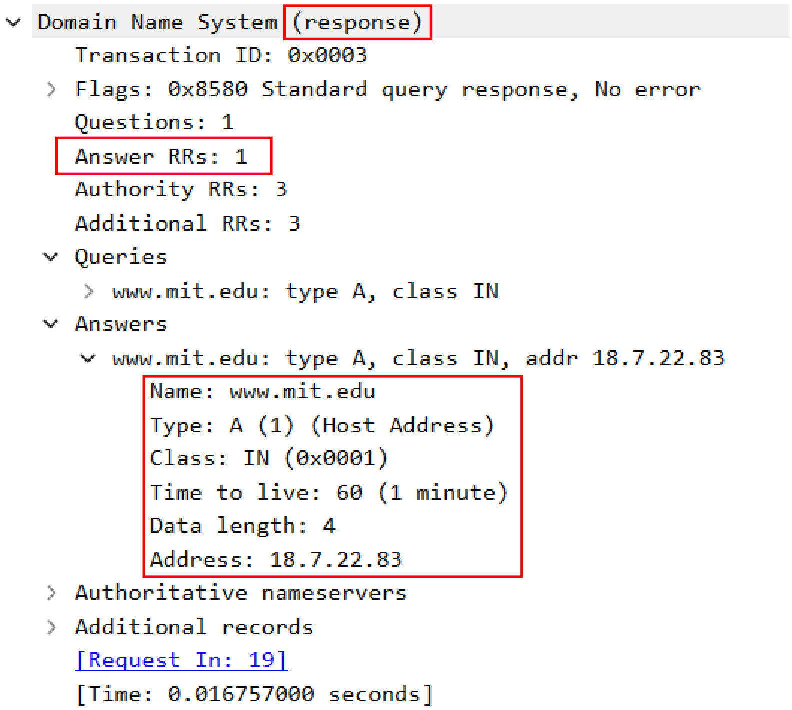


Figure C.3.10

11. Examine the DNS response message. How many “answers” are provided? What do each of these answers contain? Add screenshots in your answer.
There are 1 “answer” provided and it contains domain name, address type, class, time to live, data length and IP address.



```
v Domain Name System (response)
    Transaction ID: 0x0003
  > Flags: 0x8580 Standard query response, No error
    Questions: 1
    Answer RRs: 1
    Authority RRs: 3
    Additional RRs: 3
  v Queries
    > www.mit.edu: type A, class IN
  v Answers
    v www.mit.edu: type A, class IN, addr 18.7.22.83
        Name: www.mit.edu
        Type: A (1) (Host Address)
        Class: IN (0x0001)
        Time to live: 60 (1 minute)
        Data length: 4
        Address: 18.7.22.83
  > Authoritative nameservers
  > Additional records
    [Request In: 19]
    [Time: 0.016757000 seconds]
```

Figure C.3.11